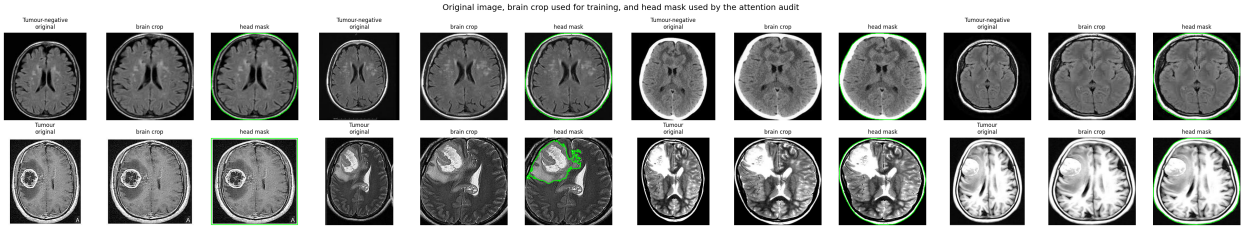
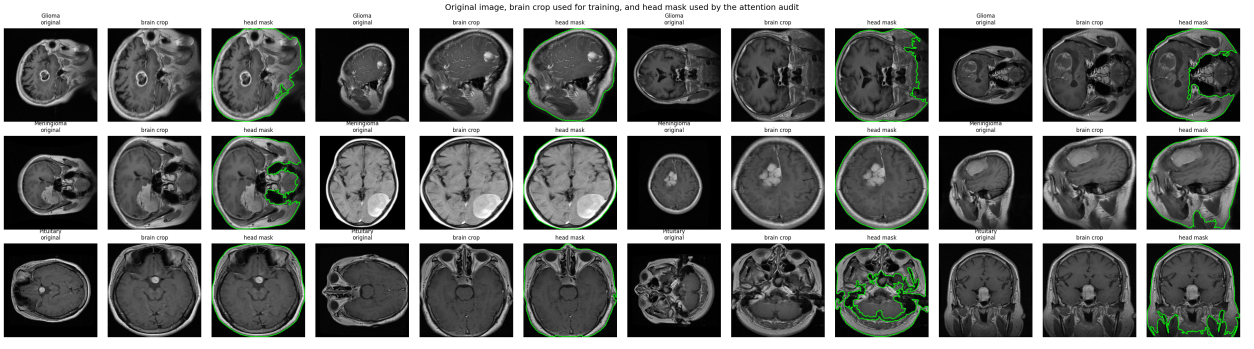
# Supplementary Material: Brain Tumour Magnetic Resonance Image Classification with Perceptual-Hash Deduplication

## S1. Data




(a) Binary detection.



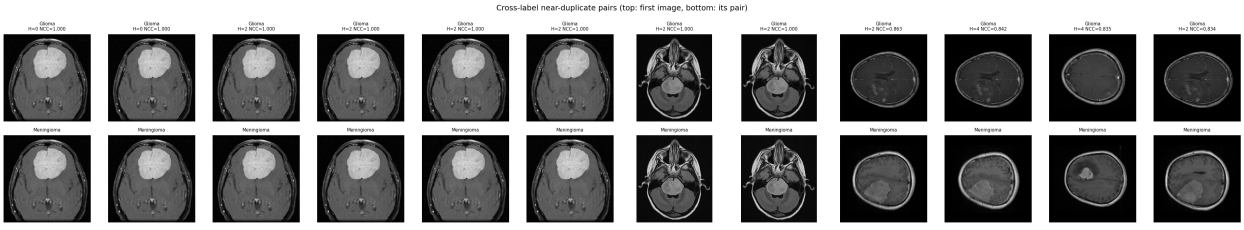

(b) Subtype classification.

**Supplementary F1:** Sample images of each class, the brain crop that every model is trained and evaluated on, and the head masks used by the attention and shortcut analyses. The MRI sequence is unknown in both pools: neither binary detection source records it, and in the subtype classification pool only the Cheng component is known to be contrast-enhanced T1-weighted, so each pool is a mixture of unknown composition and a model trained on it cannot be specified for a particular acquisition protocol.

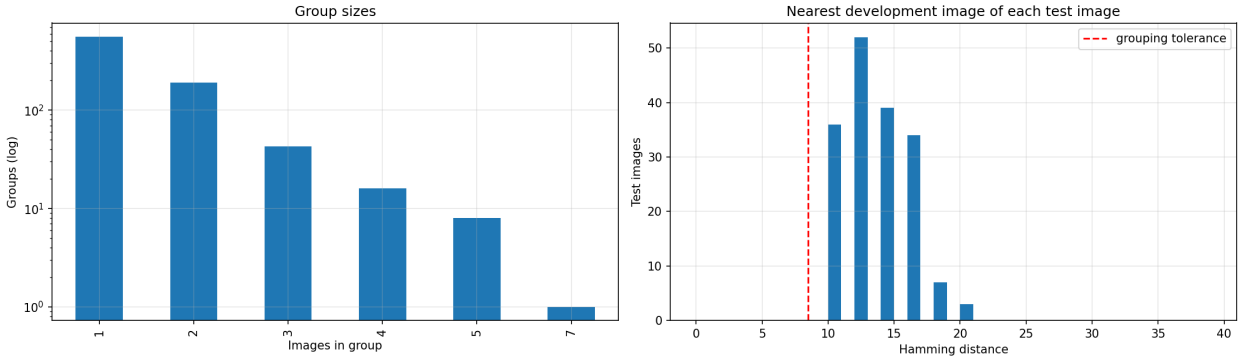
## S2. Deduplication and Splitting

**Supplementary T1:** Images kept by the label-aware perceptual-hash pass at each Hamming tolerance, with the number of image pairs of the whole pool at each exact distance. The study uses tolerance 5. Every recorded hash has exactly 32 of its 64 bits set, so every distance is even, no pair lies at an odd distance, and the tolerances paired in each column give the same pool. Pair counts were computed from the hashes stored in the per-image split table of each notebook. Cross-label pairs are image pairs with different labels. Duplicates were removed keeping the copy from the first-listed folder. Binary detection, 2,068 removals: 729 Br35H images whose retained copy is a Chakrabarty image, 1,292 duplicates within Br35H and 47 within Chakrabarty. Subtype classification, 1,094 removals: 608 within Bhuvaji, 483 within Cheng and 3 Cheng meningioma images against a Bhuvaji copy. The label-noise exclusion then brings the 4,734 images at tolerance 5 to 4,730. The per-image split table records the same pool as 1,088 duplicates and 10 label-noise exclusions (2 glioma, 8 meningioma).

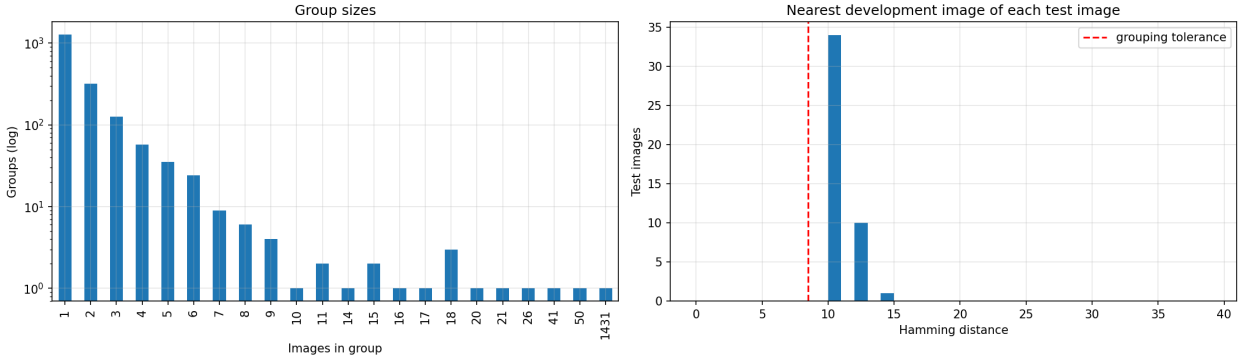
| Tolerance | 0, 1 | 2, 3 | 4, 5 | 6, 7 | 8, 9 | 10, 11 | 12 |
|---|---|---|---|---|---|---|---|
| Exact distance of pairs | 0 | 2 | 4 | 6 | 8 | 10 | 12 |
| *Binary detection* | | | | | | | |
| Tumour-negative | 860 | 583 | 455 | 395 | 341 | 310 | 285 |
| Tumour | 1,217 | 912 | 730 | 607 | 541 | 494 | 424 |
| **Total** | **2,077** | **1,495** | **1,185** | **1,002** | **882** | **804** | **709** |
| Same-label pairs | 2,299 | 2,345 | 1,754 | 1,335 | 1,069 | 1,343 | 3,993 |
| Cross-label pairs | 0 | 0 | 0 | 1 | 64 | 334 | 1,408 |
| *Subtype classification* | | | | | | | |
| Glioma | 2,303 | 2,221 | 2,010 | 1,653 | 1,277 | 919 | 627 |
| Meningioma | 1,518 | 1,453 | 1,316 | 1,130 | 935 | 704 | 506 |
| Pituitary | 1,751 | 1,663 | 1,408 | 1,145 | 886 | 663 | 478 |
| **Total** | **5,572** | **5,337** | **4,734** | **3,928** | **3,098** | **2,286** | **1,611** |
| Same-label pairs | 334 | 338 | 920 | 1,854 | 3,658 | 8,423 | 19,926 |
| Cross-label pairs | 2 | 10 | 19 | 222 | 1,116 | 4,928 | 15,895 |




**Supplementary F2:** Subtype classification cross-label pairs within Hamming distance 5, label-noise pairs first. All 31 pairs are glioma against meningioma. The 8 label-noise pairs lie inside the Bhuvaji collection with a normalised cross-correlation (NCC) of at least 0.9997, and both copies were excluded. The 23 hash false positives have NCC between 0.67 and 0.86 and were kept. The binary detection pool has no cross-label pair.

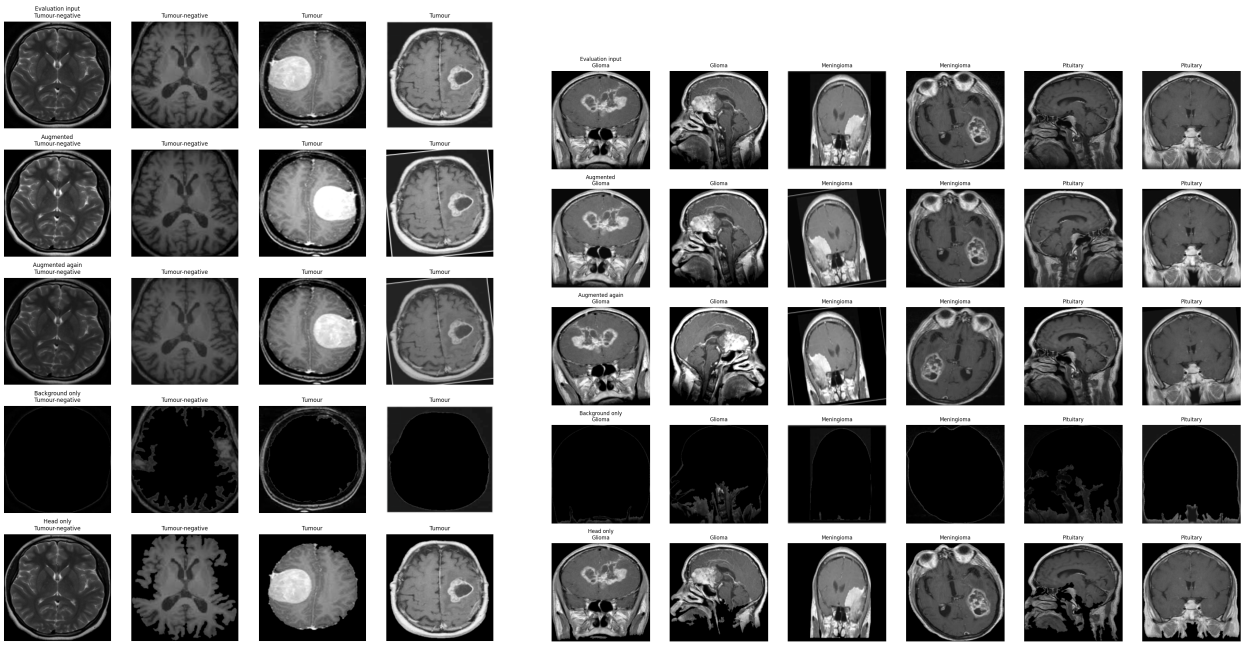



(a) Binary detection: 821 groups, the largest of 7 images.




(b) Subtype classification: 1,867 groups, the largest of 1,431 images.

**Supplementary F3:** Group sizes, and the Hamming distance and NCC from each test image to its nearest development image (leakage audit). No test image has a development image within Hamming distance 5 or 8 or at NCC $\geq 0.98$. The subtype classification group of 1,431 images holds more than 5% of the pool. The notebook warned that a smaller grouping tolerance should be considered, and the run kept tolerance 8.




(a) Binary detection.

(b) Subtype classification.

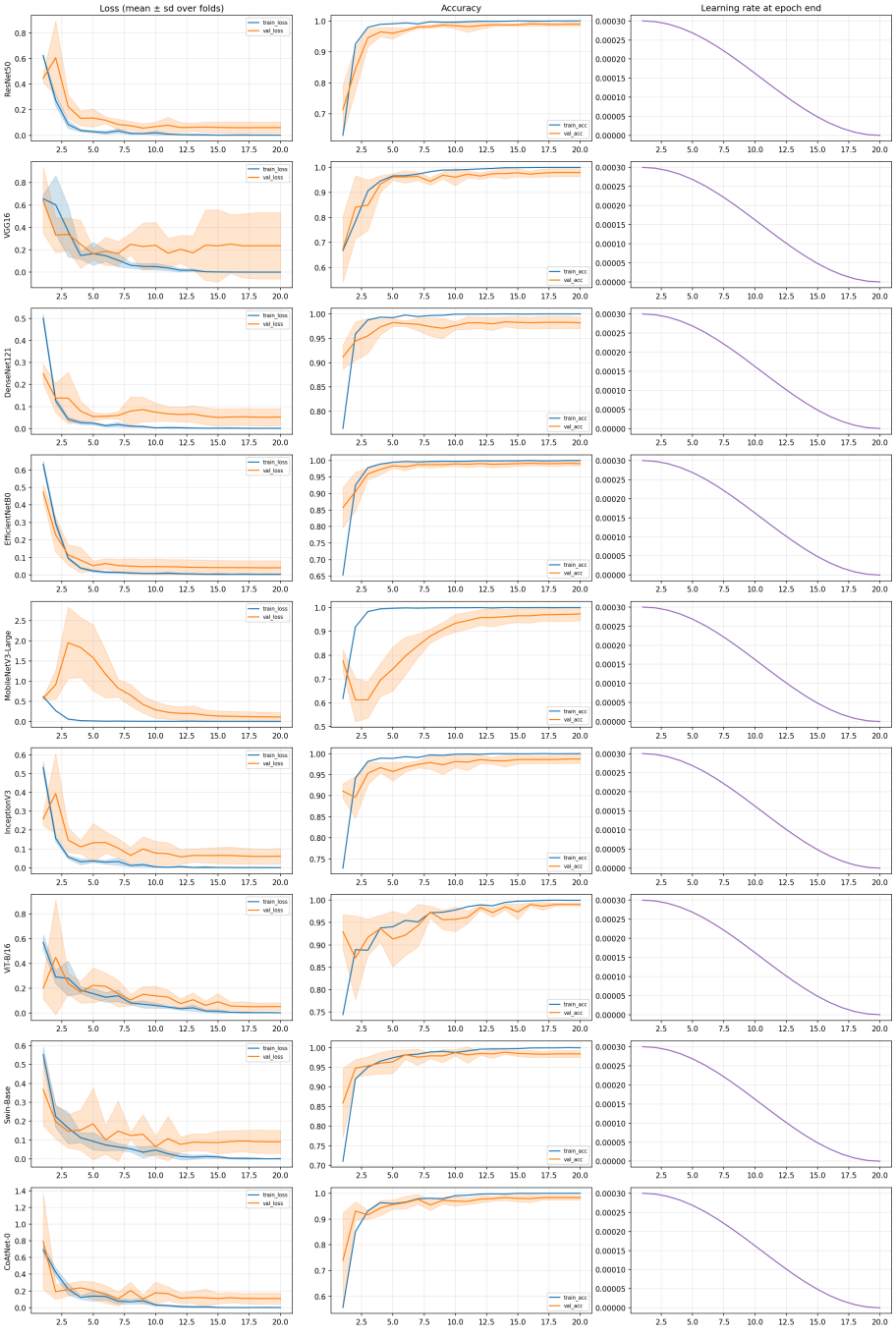
**Supplementary F4:** Training augmentation on a few training images, with the background-only and head-only inputs of the masking and shortcut tests. Augmentation runs on the GPU: horizontal flip with probability 0.5, rotation up to $\pm 10°$, and brightness and contrast jitter of up to 20%. Validation and test images are not augmented.

## S3. Training and Evaluation Details

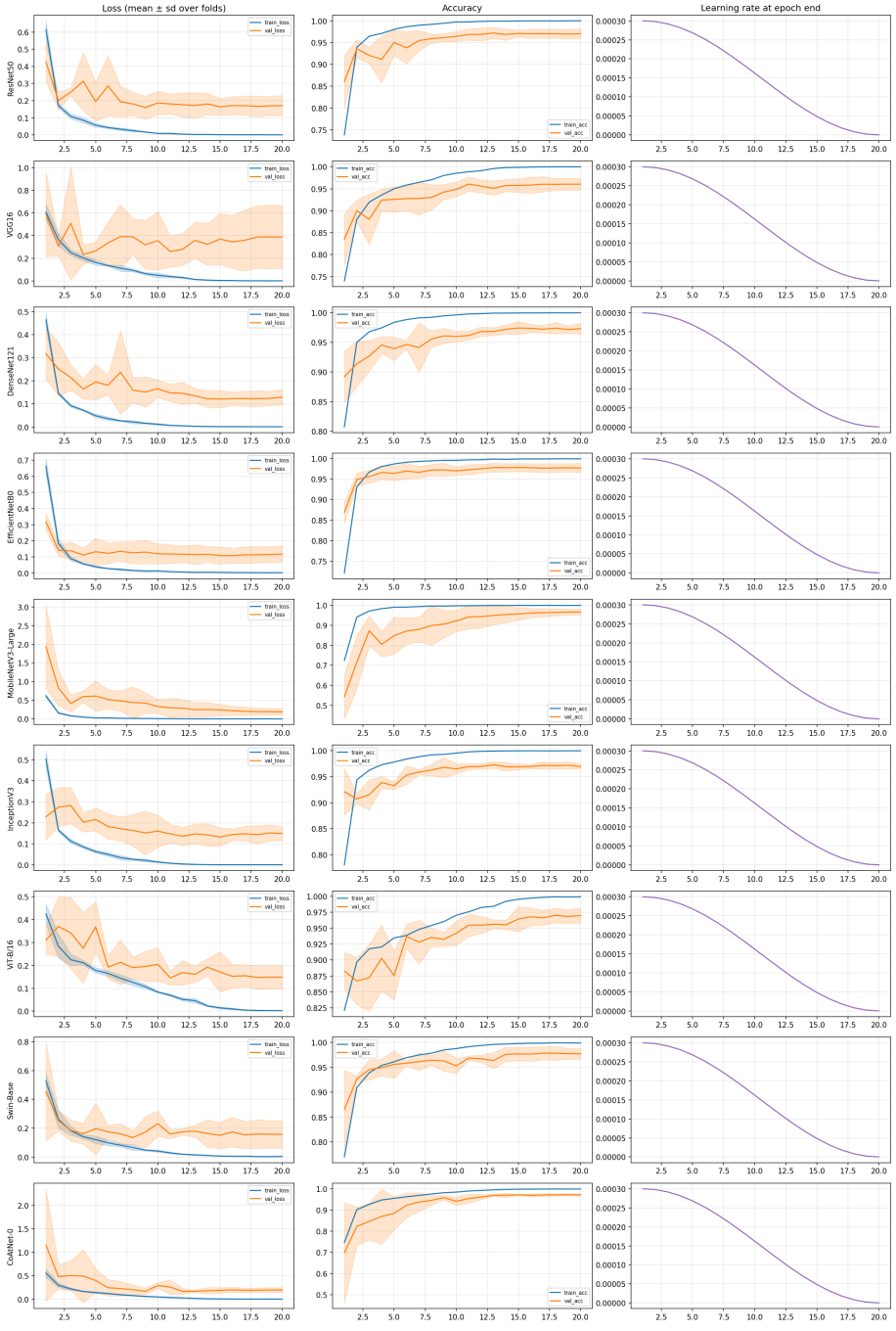
**Supplementary T2:** Training and evaluation settings, identical for all nine models and both tasks.

| Setting | Value |
|---|---|
| Hardware | Kaggle, two NVIDIA Tesla T4 GPUs, four CPU cores, 33.7 GB system memory, and one worker process per GPU training different (model, fold) jobs in parallel |
| Software | Python 3.13.15, PyTorch 2.11.0, torchvision 0.26.0, timm 1.0.29, CUDA 12.8 |
| Input | $224 \times 224$ pixels ($299 \times 299$ for InceptionV3), scaled to $[0, 1]$ and normalised with the ImageNet channel means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225) |
| Augmentation | Horizontal flip ($p = 0.5$), rotation up to $\pm 10°$, brightness and contrast jitter of 20% |
| Epochs | 20, with 1 warm-up epoch, and every training image is used in every epoch |
| Effective batch | 64 images: micro-batches of 64, or of 32 accumulated over two steps for Swin-Base, the largest that fitted on one GPU |
| Optimiser steps | Equal for every model on a given fold: 260 per binary detection run (13 per epoch), and 1,420, 1,240, 1,260, 1,200 and 780 on subtype classification folds 0 to 4, whose training sets differ in size |
| Optimiser | AdamW, learning rate $3 \times 10^{-4}$, weight decay 0.01 |
| Learning rate schedule | Linear warm-up, then cosine decay per optimiser step |
| Loss | Binary cross-entropy with logits (binary detection) and cross-entropy (subtype classification) |
| Precision | Half-precision autocasting with gradient scaling |
| Gradient clipping | Norm 1.0 |
| Checkpoint selection | Lowest validation log-loss (a strictly proper scoring rule), evaluated after every epoch |
| Cross-validation | 5 group-aware folds plus a fixed held-out test set |
| Runs per task | 45 full-image runs and 9 background-only runs, all 54 completed, with 63.5 minutes of training for binary detection and 264.6 for subtype classification |
| Random seed | 42, with deterministic kernels not enforced, so reruns reproduce the figures closely but not bit for bit |
| Calibration | Temperature fitted per run on its validation fold over $\log T \in [-3, 3]$, with binary detection reported at threshold 0.5 and at the mean validation-tuned Youden threshold |
| Uncertainty | Bootstrap 95% confidence intervals of accuracy and of the accuracy gap to the leading model. A gap interval that includes zero cannot separate a model from the leader |
| Attention maps | Grad-CAM at the last spatial or token stage (the last convolution for VGG16) of the fold model with the lowest validation loss. Occlusion sensitivity computed identically for every model on a common stratified test sample, because Grad-CAM maps of convolutional and attention stages are not strictly commensurable |
| Efficiency | Latency over 100 single-image passes after 10 warm-up passes, and throughput at batch size 32 over 20 passes after 3 warm-up passes, each repeated three times. Training time, throughput and peak memory averaged over the folds |

## S4. Cross-Validation Details




**Supplementary F5:** Binary detection: mean ± one standard deviation across folds of training and validation loss, validation accuracy and learning rate, per model.




**Supplementary F6:** Subtype classification: mean ± one standard deviation across folds of training and validation loss, validation accuracy and learning rate, per model.

**Supplementary T3:** Binary detection: every full-image run. Every run took 260 optimiser steps. $T$ is the temperature fitted on the run's validation fold and Thr. the Youden threshold chosen there. Accuracy and log-loss are on the 171 test images after temperature scaling. In fold 4 the temperature of six models lies at or near the lower search bound $e^{-3} \approx 0.0498$.

| Model | Fold | Best epoch | $T$ | Thr. | Test accuracy | Test log-loss | Train images |
|---|---|---|---|---|---|---|---|
| CoAtNet-0 | 0 | 9 | 1.1767 | 0.739 | 0.9883 | 0.0491 | 830 |
| | 1 | 7 | 1.4761 | 0.529 | 0.9825 | 0.0534 | 803 |
| | 2 | 12 | 1.7756 | 0.633 | 0.9474 | 0.1317 | 807 |
| | 3 | 14 | 2.1222 | 0.316 | 0.9883 | 0.0333 | 804 |
| | 4 | 7 | 0.0604 | 1.000 | 0.9708 | 0.8079 | 812 |
| DenseNet121 | 0 | 6 | 0.8003 | 0.724 | 0.9766 | 0.0847 | 830 |
| | 1 | 11 | 1.0106 | 0.657 | 0.9649 | 0.0751 | 803 |
| | 2 | 12 | 1.3466 | 0.885 | 0.9649 | 0.1019 | 807 |
| | 3 | 14 | 1.1351 | 0.950 | 0.9825 | 0.0696 | 804 |
| | 4 | 15 | 0.5882 | 0.927 | 0.9708 | 0.2513 | 812 |
| EfficientNetB0 | 0 | 5 | 1.2527 | 0.621 | 0.9708 | 0.0875 | 830 |
| | 1 | 14 | 1.2311 | 0.962 | 0.9766 | 0.0378 | 803 |
| | 2 | 13 | 1.1739 | 0.625 | 0.9825 | 0.0471 | 807 |
| | 3 | 11 | 0.9215 | 0.985 | 0.9825 | 0.0755 | 804 |
| | 4 | 19 | 0.0498 | 1.000 | 0.9766 | 0.5431 | 812 |
| InceptionV3 | 0 | 17 | 1.6119 | 0.895 | 0.9766 | 0.0543 | 830 |
| | 1 | 8 | 1.4656 | 0.045 | 0.9766 | 0.0603 | 803 |
| | 2 | 10 | 0.9470 | 0.861 | 0.9766 | 0.1005 | 807 |
| | 3 | 5 | 1.1069 | 0.829 | 0.9825 | 0.0372 | 804 |
| | 4 | 12 | 0.0682 | 1.000 | 0.9825 | 0.3662 | 812 |
| MobileNetV3-Large | 0 | 20 | 2.7638 | 0.121 | 0.9298 | 0.1551 | 830 |
| | 1 | 20 | 1.8160 | 0.583 | 0.9766 | 0.0638 | 803 |
| | 2 | 14 | 1.6694 | 0.410 | 0.9883 | 0.0448 | 807 |
| | 3 | 20 | 1.7832 | 0.249 | 0.9591 | 0.1161 | 804 |
| | 4 | 13 | 0.0508 | 1.000 | 0.9649 | 0.7224 | 812 |
| ResNet50 | 0 | 9 | 1.7422 | 0.819 | 0.9474 | 0.1451 | 830 |
| | 1 | 8 | 1.7905 | 0.314 | 0.9766 | 0.0837 | 803 |
| | 2 | 7 | 1.0362 | 0.853 | 0.9649 | 0.0787 | 807 |
| | 3 | 17 | 1.2096 | 0.759 | 0.9708 | 0.1416 | 804 |
| | 4 | 12 | 0.0498 | 1.000 | 0.9825 | 0.4848 | 812 |
| Swin-Base | 0 | 14 | 1.7903 | 0.617 | 0.9708 | 0.0826 | 830 |
| | 1 | 11 | 1.7026 | 0.550 | 0.9708 | 0.1311 | 803 |
| | 2 | 6 | 1.5159 | 0.683 | 0.9591 | 0.1221 | 807 |
| | 3 | 5 | 0.8935 | 0.261 | 0.9883 | 0.0536 | 804 |
| | 4 | 10 | 0.0498 | 1.000 | 1.0000 | 0.0000 | 812 |
| VGG16 | 0 | 5 | 1.7926 | 0.462 | 0.9532 | 0.1252 | 830 |
| | 1 | 13 | 2.1367 | 0.216 | 0.9649 | 0.0912 | 803 |
| | 2 | 18 | 1.5436 | 0.775 | 0.9649 | 0.2265 | 807 |
| | 3 | 4 | 1.4973 | 0.196 | 0.9474 | 0.1276 | 804 |
| | 4 | 6 | 1.8620 | 0.495 | 0.9649 | 0.1407 | 812 |
| ViT-B/16 | 0 | 17 | 1.4451 | 0.767 | 0.9825 | 0.0566 | 830 |
| | 1 | 12 | 1.5874 | 0.483 | 0.9649 | 0.1114 | 803 |
| | 2 | 17 | 1.3066 | 0.824 | 0.9883 | 0.0526 | 807 |
| | 3 | 8 | 0.8068 | 0.559 | 0.9708 | 0.0711 | 804 |
| | 4 | 16 | 0.9993 | 0.836 | 0.9708 | 0.1007 | 812 |

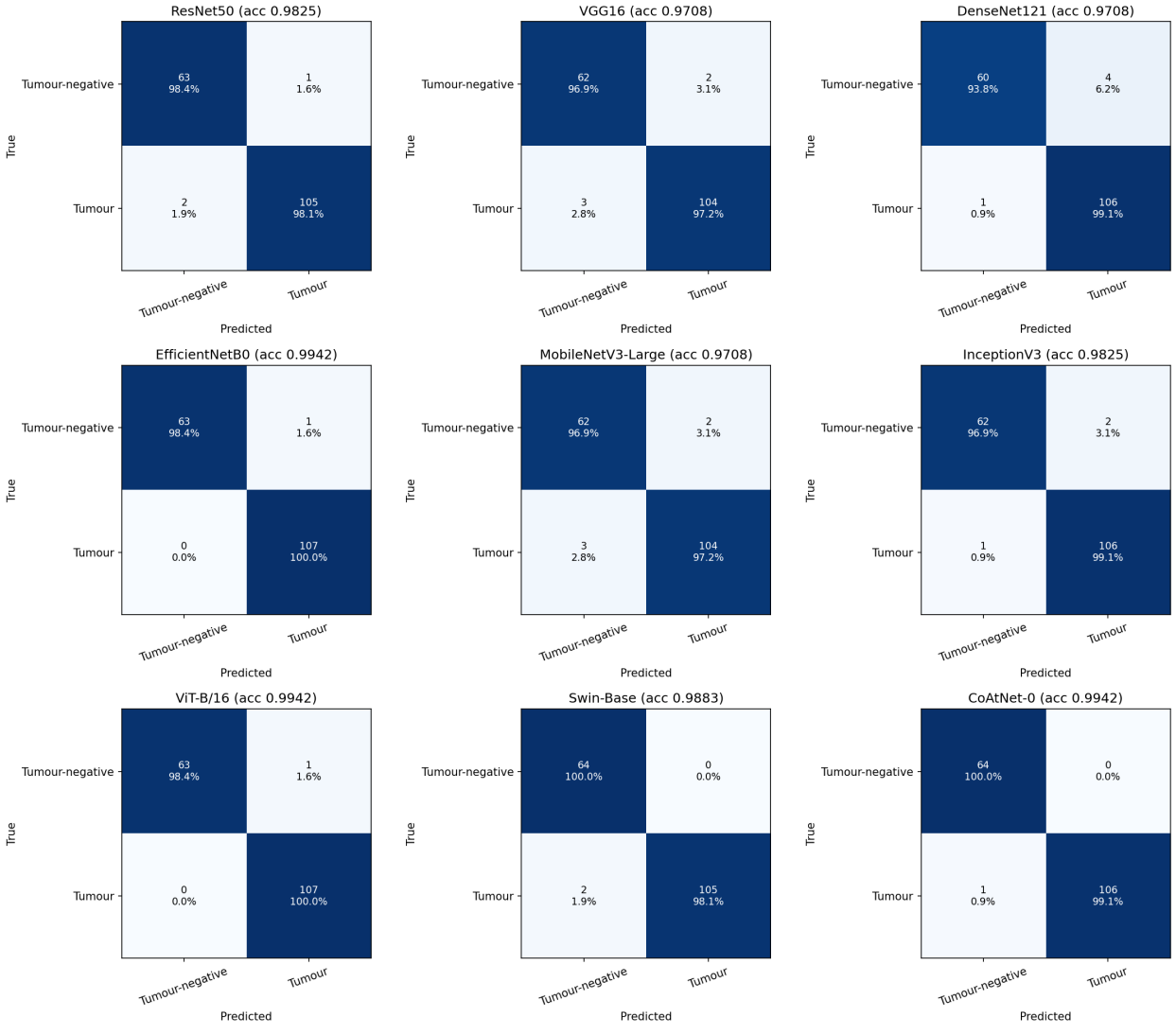
**Supplementary T4:** Subtype classification: every full-image run. Optimiser steps depend on the fold because the group-aware folds differ in size, but within a fold they are equal for every model. In fold 0, whose validation fold has 154 images, the temperature of EfficientNetB0, InceptionV3 and Swin-Base lies at or near the lower search bound.

| Model | Fold | Steps | Best epoch | $T$ | Test accuracy | Test log-loss | Train / val images |
|---|---|---|---|---|---|---|---|
| CoAtNet-0 | 0 | 1,420 | 7 | 0.8899 | 1.0000 | 0.0209 | 4,531 / 154 |
| | 1 | 1,240 | 9 | 1.3042 | 1.0000 | 0.0034 | 3,943 / 742 |
| | 2 | 1,260 | 5 | 0.8503 | 0.9778 | 0.0720 | 3,988 / 697 |
| | 3 | 1,200 | 6 | 1.2193 | 0.9778 | 0.0395 | 3,809 / 876 |
| | 4 | 780 | 15 | 2.3073 | 0.9778 | 0.0236 | 2,469 / 2,216 |
| DenseNet121 | 0 | 1,420 | 7 | 1.1754 | 1.0000 | 0.0040 | 4,531 / 154 |
| | 1 | 1,240 | 15 | 1.9404 | 1.0000 | 0.0061 | 3,943 / 742 |
| | 2 | 1,260 | 9 | 1.5745 | 1.0000 | 0.0186 | 3,988 / 697 |
| | 3 | 1,200 | 4 | 1.4594 | 0.9778 | 0.1067 | 3,809 / 876 |
| | 4 | 780 | 10 | 1.5314 | 0.9778 | 0.0355 | 2,469 / 2,216 |
| EfficientNetB0 | 0 | 1,420 | 9 | 0.0498 | 0.9778 | 0.6140 | 4,531 / 154 |
| | 1 | 1,240 | 4 | 1.5348 | 0.9111 | 0.1585 | 3,943 / 742 |
| | 2 | 1,260 | 15 | 1.8186 | 1.0000 | 0.0024 | 3,988 / 697 |
| | 3 | 1,200 | 3 | 1.3621 | 1.0000 | 0.0181 | 3,809 / 876 |
| | 4 | 780 | 16 | 1.7429 | 0.9778 | 0.0330 | 2,469 / 2,216 |
| InceptionV3 | 0 | 1,420 | 9 | 0.0515 | 1.0000 | 0.0000 | 4,531 / 154 |
| | 1 | 1,240 | 7 | 1.5646 | 1.0000 | 0.0050 | 3,943 / 742 |
| | 2 | 1,260 | 10 | 1.6805 | 1.0000 | 0.0040 | 3,988 / 697 |
| | 3 | 1,200 | 15 | 2.2675 | 1.0000 | 0.0101 | 3,809 / 876 |
| | 4 | 780 | 11 | 1.7383 | 0.9778 | 0.0503 | 2,469 / 2,216 |
| MobileNetV3-Large | 0 | 1,420 | 12 | 1.0519 | 0.9778 | 0.1369 | 4,531 / 154 |
| | 1 | 1,240 | 20 | 3.1578 | 0.9778 | 0.0712 | 3,943 / 742 |
| | 2 | 1,260 | 18 | 2.6818 | 0.9778 | 0.0685 | 3,988 / 697 |
| | 3 | 1,200 | 8 | 2.4891 | 0.9778 | 0.0327 | 3,809 / 876 |
| | 4 | 780 | 20 | 3.0036 | 0.9778 | 0.0689 | 2,469 / 2,216 |
| ResNet50 | 0 | 1,420 | 5 | 0.9164 | 1.0000 | 0.0026 | 4,531 / 154 |
| | 1 | 1,240 | 9 | 2.0543 | 0.9778 | 0.0395 | 3,943 / 742 |
| | 2 | 1,260 | 5 | 1.5465 | 0.9778 | 0.0672 | 3,988 / 697 |
| | 3 | 1,200 | 15 | 2.6184 | 1.0000 | 0.0176 | 3,809 / 876 |
| | 4 | 780 | 8 | 1.6629 | 0.9778 | 0.0957 | 2,469 / 2,216 |
| Swin-Base | 0 | 1,420 | 15 | 0.0498 | 1.0000 | 0.0000 | 4,531 / 154 |
| | 1 | 1,240 | 8 | 1.6375 | 1.0000 | 0.0112 | 3,943 / 742 |
| | 2 | 1,260 | 9 | 1.9777 | 1.0000 | 0.0076 | 3,988 / 697 |
| | 3 | 1,200 | 5 | 1.7771 | 0.9778 | 0.0397 | 3,809 / 876 |
| | 4 | 780 | 18 | 2.2403 | 1.0000 | 0.0047 | 2,469 / 2,216 |
| VGG16 | 0 | 1,420 | 11 | 1.2077 | 0.9778 | 0.1367 | 4,531 / 154 |
| | 1 | 1,240 | 7 | 1.7136 | 1.0000 | 0.0175 | 3,943 / 742 |
| | 2 | 1,260 | 2 | 1.6416 | 0.9333 | 0.2497 | 3,988 / 697 |
| | 3 | 1,200 | 4 | 2.0744 | 0.9778 | 0.0520 | 3,809 / 876 |
| | 4 | 780 | 13 | 2.2774 | 0.9778 | 0.0970 | 2,469 / 2,216 |
| ViT-B/16 | 0 | 1,420 | 15 | 1.3910 | 1.0000 | 0.0020 | 4,531 / 154 |
| | 1 | 1,240 | 13 | 1.7649 | 1.0000 | 0.0238 | 3,943 / 742 |
| | 2 | 1,260 | 13 | 1.2801 | 0.9778 | 0.0420 | 3,988 / 697 |
| | 3 | 1,200 | 11 | 1.5805 | 1.0000 | 0.0129 | 3,809 / 876 |
| | 4 | 780 | 11 | 1.4512 | 0.9778 | 0.1176 | 2,469 / 2,216 |

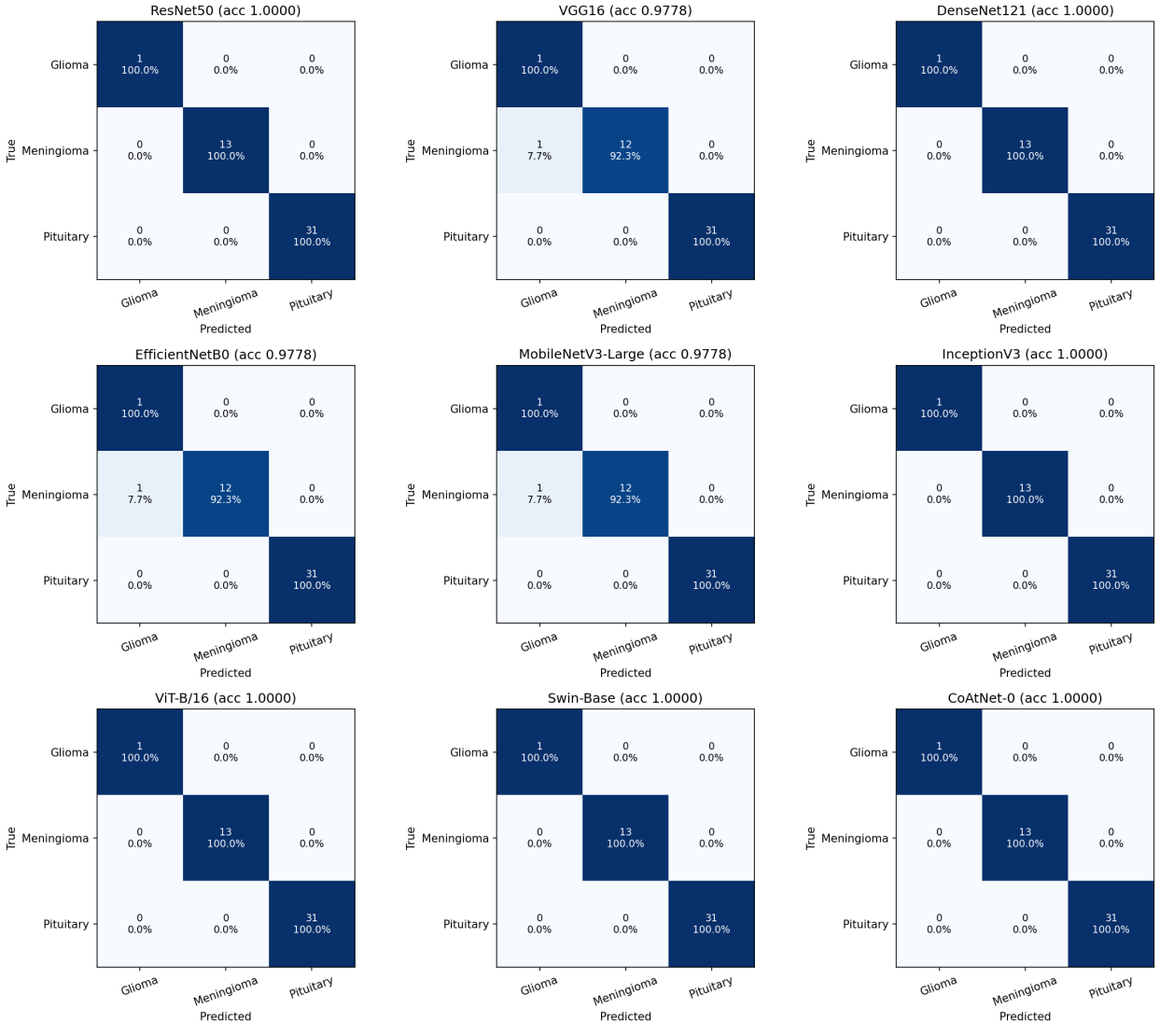
## S5. Fold-Ensemble Test Results

**Supplementary T5:** Binary detection: fold ensembles on the 171 held-out test images (107 tumour, 64 tumour-negative). Sensitivity is recall of the tumour class and specificity recall of the tumour-negative class, both at threshold 0.5. FN and FP are missed tumours and false alarms. Accuracy has a bootstrap 95% confidence interval. Rows are ordered by accuracy. At the mean validation-tuned Youden thresholds, from 0.43 to 0.84, accuracy lies between 96.49% and 98.83%: the tuned thresholds raise it only for DenseNet121 (97.66%), leave it unchanged for InceptionV3 and VGG16 and lower it for the other six models, and EfficientNetB0 sensitivity falls to 96.26% with specificity unchanged at 98.44%.

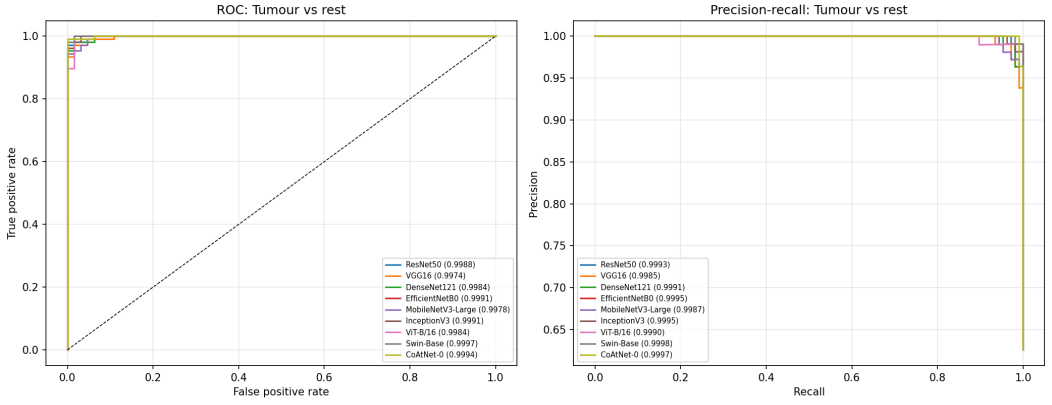
| Model | Accuracy (%) [95% CI] | Bal. acc. (%) | Sens. (%) | Spec. (%) | MCC | AUC | FN | FP |
|---|---|---|---|---|---|---|---|---|
| CoAtNet-0 | 99.42 [98.25, 100] | 99.53 | 99.07 | 100.00 | 0.9876 | 0.9994 | 1 | 0 |
| EfficientNetB0 | 99.42 [98.25, 100] | 99.22 | 100.00 | 98.44 | 0.9876 | 0.9991 | 0 | 1 |
| ViT-B/16 | 99.42 [98.25, 100] | 99.22 | 100.00 | 98.44 | 0.9876 | 0.9984 | 0 | 1 |
| Swin-Base | 98.83 [97.08, 100] | 99.07 | 98.13 | 100.00 | 0.9755 | 0.9997 | 2 | 0 |
| InceptionV3 | 98.25 [95.91, 100] | 97.97 | 99.07 | 96.88 | 0.9625 | 0.9991 | 1 | 2 |
| ResNet50 | 98.25 [95.91, 100] | 98.28 | 98.13 | 98.44 | 0.9627 | 0.9988 | 2 | 1 |
| DenseNet121 | 97.08 [94.15, 99.42] | 96.41 | 99.07 | 93.75 | 0.9376 | 0.9984 | 1 | 4 |
| MobileNetV3-Large | 97.08 [94.15, 99.42] | 97.04 | 97.20 | 96.88 | 0.9378 | 0.9978 | 3 | 2 |
| VGG16 | 97.08 [94.15, 99.42] | 97.04 | 97.20 | 96.88 | 0.9378 | 0.9974 | 3 | 2 |



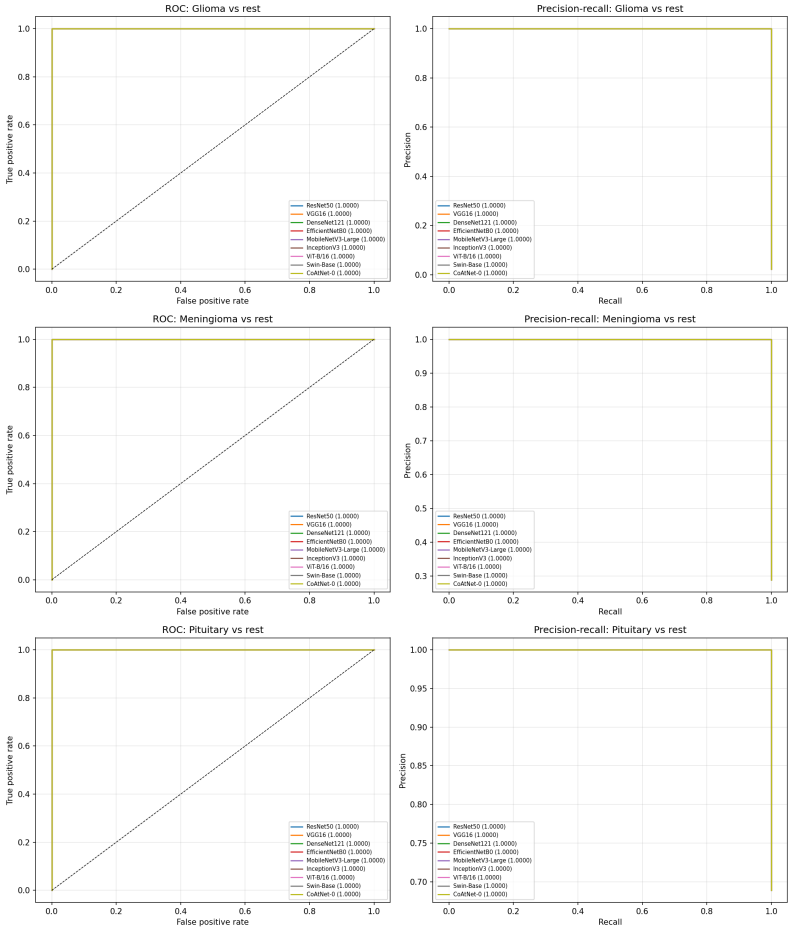

**Supplementary F7:** Binary detection: test confusion matrices of the fold ensemble of every model, as counts with row percentages.




**Supplementary F8:** Subtype classification: test confusion matrices of every fold ensemble, as counts with row percentages.

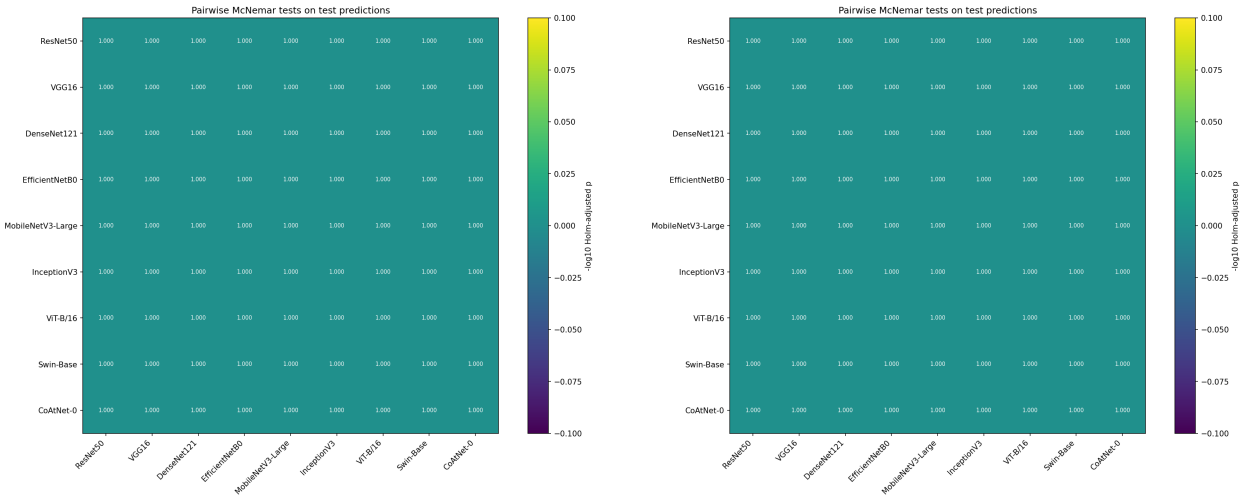



(a) Binary detection, tumour class.




(b) Subtype classification, one-versus-rest.

**Supplementary F9:** Receiver operating characteristic and precision–recall curves of every fold ensemble on the test set.

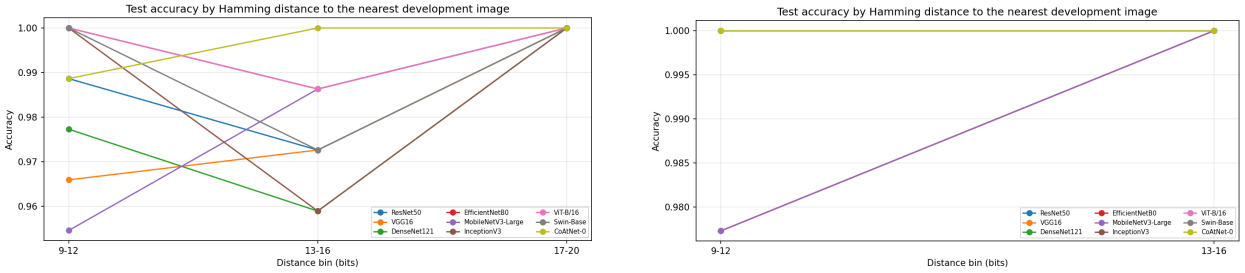



(a) Binary detection. (b) Subtype classification.

**Supplementary F10:** Holm-corrected exact McNemar p-values between the fold ensembles. Every adjusted p-value is 1.0 in both tasks.

**Supplementary T6:** Fold-ensemble test accuracy by the Hamming distance from each test image to its nearest development image. The subtype classification test set has no image at distance 17 to 20.

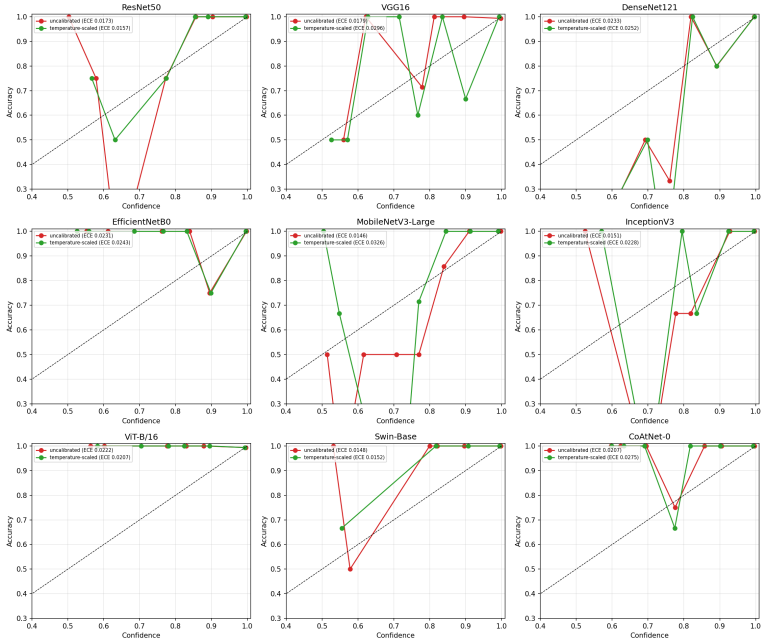
| | Binary detection | | | Subtype classification | |
|---|---|---|---|---|---|
| Model | 9–12 | 13–16 | 17–20 | 9–12 | 13–16 |
| CoAtNet-0 | 0.9886 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| DenseNet121 | 0.9773 | 0.9589 | 1.0000 | 1.0000 | 1.0000 |
| EfficientNetB0 | 1.0000 | 0.9863 | 1.0000 | 0.9773 | 1.0000 |
| InceptionV3 | 1.0000 | 0.9589 | 1.0000 | 1.0000 | 1.0000 |
| MobileNetV3-Large | 0.9545 | 0.9863 | 1.0000 | 0.9773 | 1.0000 |
| ResNet50 | 0.9886 | 0.9726 | 1.0000 | 1.0000 | 1.0000 |
| Swin-Base | 1.0000 | 0.9726 | 1.0000 | 1.0000 | 1.0000 |
| VGG16 | 0.9659 | 0.9726 | 1.0000 | 0.9773 | 1.0000 |
| ViT-B/16 | 1.0000 | 0.9863 | 1.0000 | 1.0000 | 1.0000 |



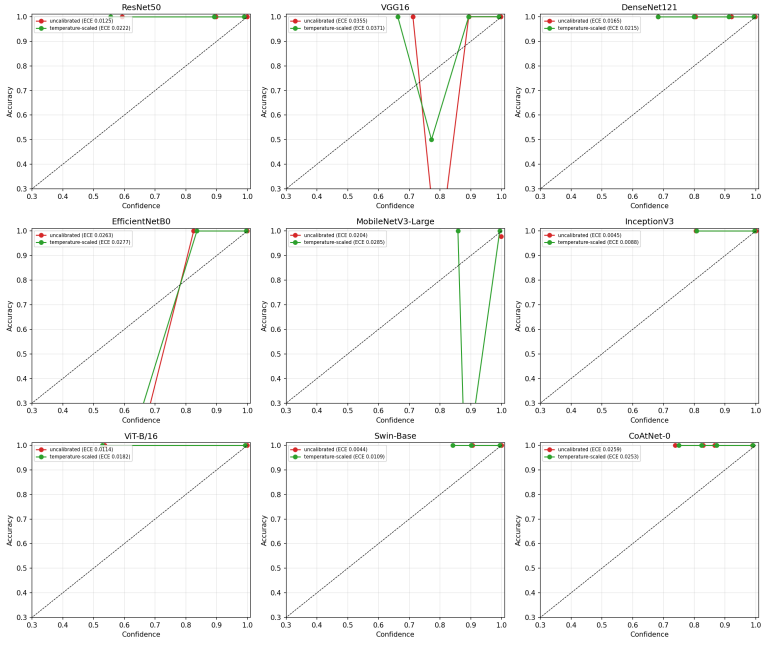

(a) Binary detection. (b) Subtype classification.

**Supplementary F11:** Fold-ensemble test accuracy by the Hamming distance from each test image to its nearest development image, the values of Supplementary T6.

## S6. Calibration and Efficiency




(a) Binary detection.

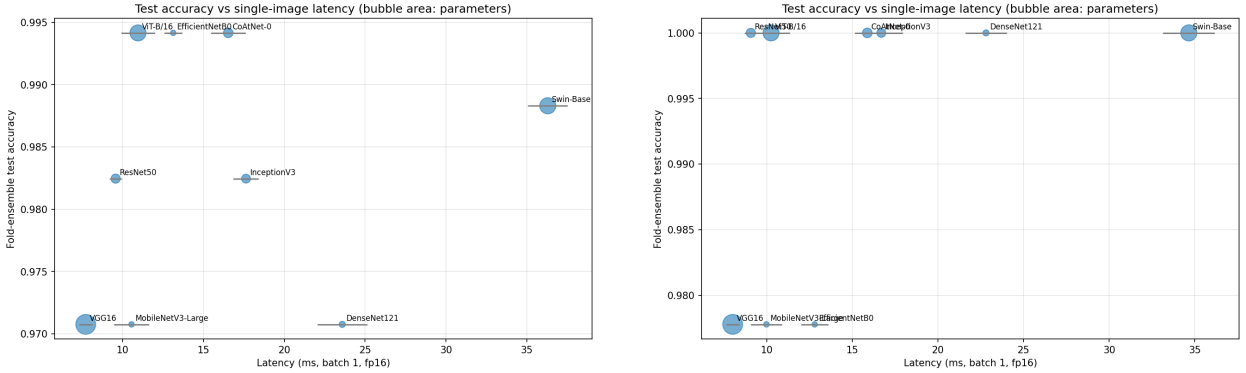



(b) Subtype classification.

**Supplementary F12:** Reliability diagrams (accuracy against confidence in each confidence bin) of every fold ensemble on the test set, uncalibrated and temperature-scaled, with the expected calibration error of each.

**Supplementary T7:** Computational cost of the nine backbones. Bin. and Sub. denote binary detection and subtype classification. Parameters and GFLOPs are measured on the binary detection models at the native input ($299 \times 299$ for InceptionV3, which needs 5.67 GFLOPs at $224 \times 224$). Latency is the mean of single-image passes and throughput the mean at batch size 32, both on one Tesla T4 under half-precision autocasting and pooled over three repeats. Training time is minutes per run, averaged over folds. Peak training memory ranges from 2.1 GB (MobileNetV3-Large) to 11.4 GB (CoAtNet-0). Rows are ordered by parameter count.

| | | | Latency (ms) | | Throughput (img/s) | | Training (min/run) | |
|---|---|---|---|---|---|---|---|---|
| Model | Params (M) | GFLOPs | Bin. | Sub. | Bin. | Sub. | Bin. | Sub. |
| EfficientNetB0 | 4.01 | 0.77 | 13.1 | 12.8 | 1,393 | 1,266 | 2.0 | 5.1 |
| MobileNetV3-Large | 4.20 | 0.43 | 10.6 | 10.0 | 2,301 | 2,183 | 1.5 | 3.5 |
| DenseNet121 | 6.95 | 5.67 | 23.6 | 22.8 | 657 | 608 | 1.5 | 6.9 |
| InceptionV3 | 21.79 | 11.42 | 17.6 | 16.7 | 512 | 438 | 1.7 | 8.0 |
| ResNet50 | 23.51 | 8.17 | 9.6 | 9.1 | 777 | 717 | 1.2 | 5.3 |
| CoAtNet-0 | 26.67 | 8.42 | 16.5 | 15.9 | 513 | 448 | 2.3 | 9.5 |
| ViT-B/16 | 85.80 | 22.54 | 11.0 | 10.3 | 272 | 241 | 2.9 | 15.0 |
| Swin-Base | 86.74 | 30.86 | 36.3 | 34.6 | 198 | 173 | 4.3 | 21.4 |
| VGG16 | 134.26 | 30.93 | 7.7 | 8.0 | 386 | 332 | 2.2 | 9.5 |




(a) Binary detection.

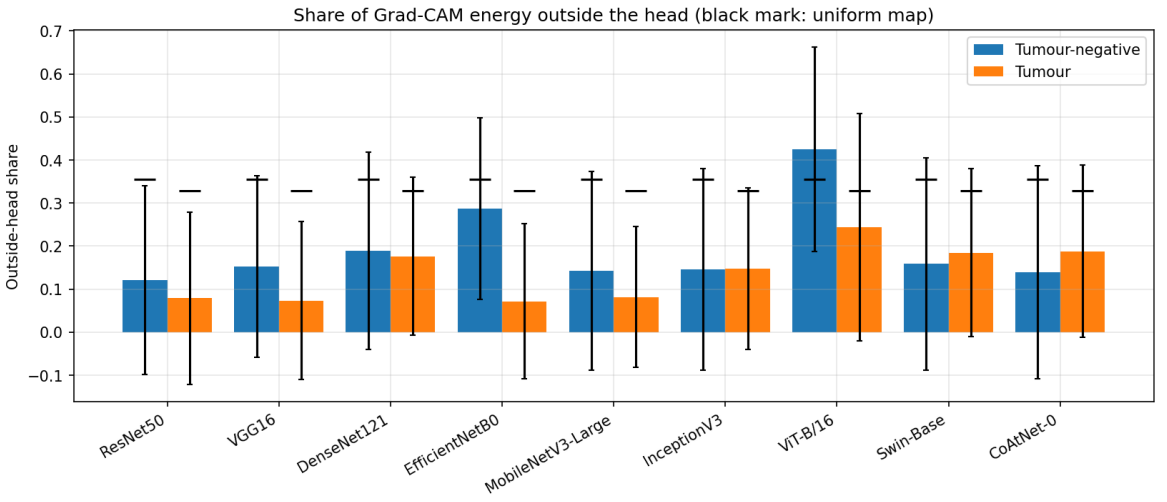
(b) Subtype classification.

**Supplementary F13:** Fold-ensemble test accuracy against single-image latency for every model.

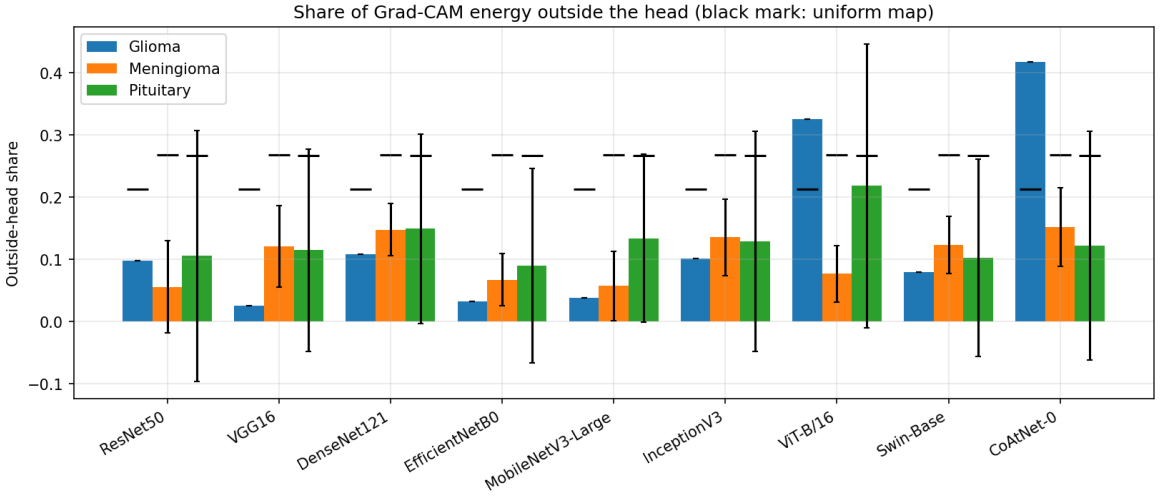
## S7. Interpretability and Shortcut Details

**Supplementary T8:** Share of Grad-CAM energy outside the head mask, per class, mean ± standard deviation over test images, and of occlusion sensitivity over 16 binary detection and 17 subtype classification test images. The subtype classification test set has one glioma image, so that column has no standard deviation.

| | Binary detection | | | Subtype classification | | | |
|---|---|---|---|---|---|---|---|
| Model | Tumour-negative | Tumour | Occlusion | Glioma | Meningioma | Pituitary | Occlusion |
| ResNet50 | $0.122 \pm 0.219$ | $0.079 \pm 0.200$ | 0.116 | 0.098 | $0.056 \pm 0.074$ | $0.106 \pm 0.202$ | 0.079 |
| VGG16 | $0.153 \pm 0.212$ | $0.074 \pm 0.184$ | 0.007 | 0.026 | $0.121 \pm 0.066$ | $0.115 \pm 0.163$ | 0.079 |
| DenseNet121 | $0.190 \pm 0.229$ | $0.176 \pm 0.183$ | 0.135 | 0.108 | $0.148 \pm 0.042$ | $0.150 \pm 0.153$ | 0.110 |
| EfficientNetB0 | $0.287 \pm 0.211$ | $0.072 \pm 0.180$ | 0.194 | 0.033 | $0.067 \pm 0.042$ | $0.091 \pm 0.156$ | 0.066 |
| MobileNetV3-Large | $0.142 \pm 0.231$ | $0.082 \pm 0.163$ | 0.111 | 0.038 | $0.058 \pm 0.056$ | $0.134 \pm 0.136$ | 0.071 |
| InceptionV3 | $0.146 \pm 0.234$ | $0.147 \pm 0.187$ | 0.167 | 0.102 | $0.136 \pm 0.062$ | $0.129 \pm 0.178$ | 0.099 |
| ViT-B/16 (Grad-CAM) | $0.426 \pm 0.238$ | $0.244 \pm 0.264$ | 0.155 | 0.326 | $0.077 \pm 0.045$ | $0.218 \pm 0.229$ | 0.107 |
| ViT-B/16 (rollout) | $0.421 \pm 0.179$ | $0.285 \pm 0.163$ | | 0.262 | $0.350 \pm 0.047$ | $0.358 \pm 0.121$ | |
| Swin-Base | $0.159 \pm 0.246$ | $0.185 \pm 0.195$ | 0.182 | 0.079 | $0.123 \pm 0.046$ | $0.103 \pm 0.158$ | 0.122 |
| CoAtNet-0 | $0.139 \pm 0.248$ | $0.188 \pm 0.200$ | 0.089 | 0.417 | $0.152 \pm 0.064$ | $0.122 \pm 0.184$ | 0.124 |
| Uniform map | 0.356 | 0.328 | | 0.213 | 0.268 | 0.267 | |

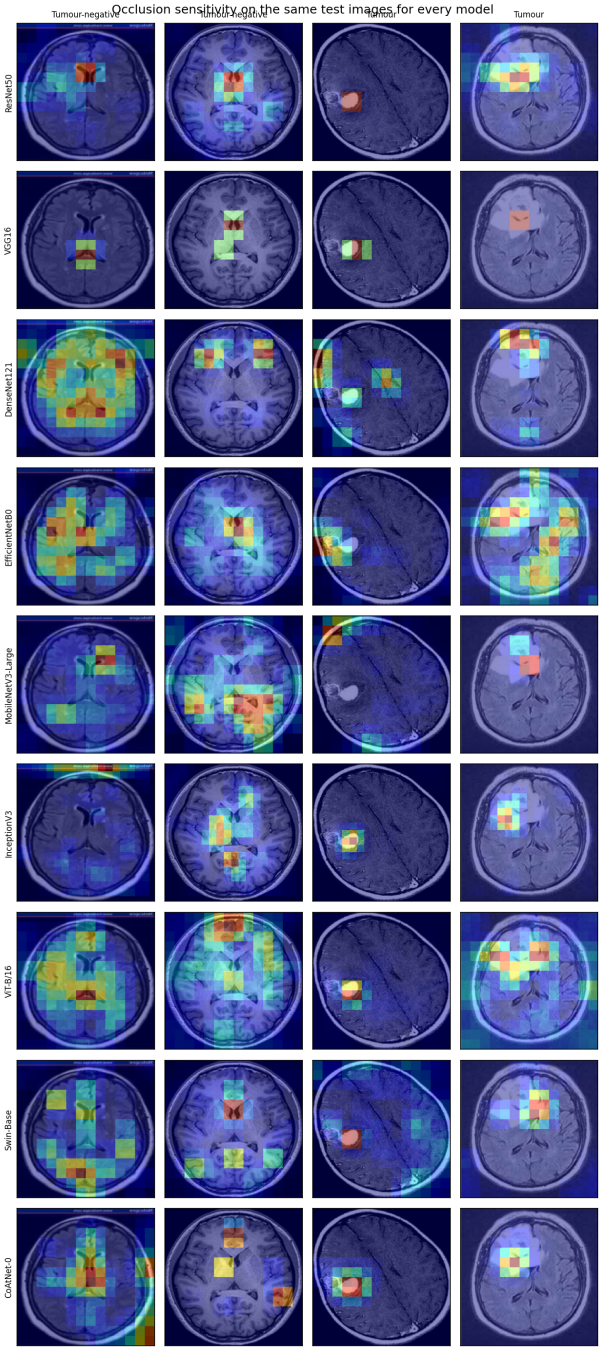



(a) Binary detection.

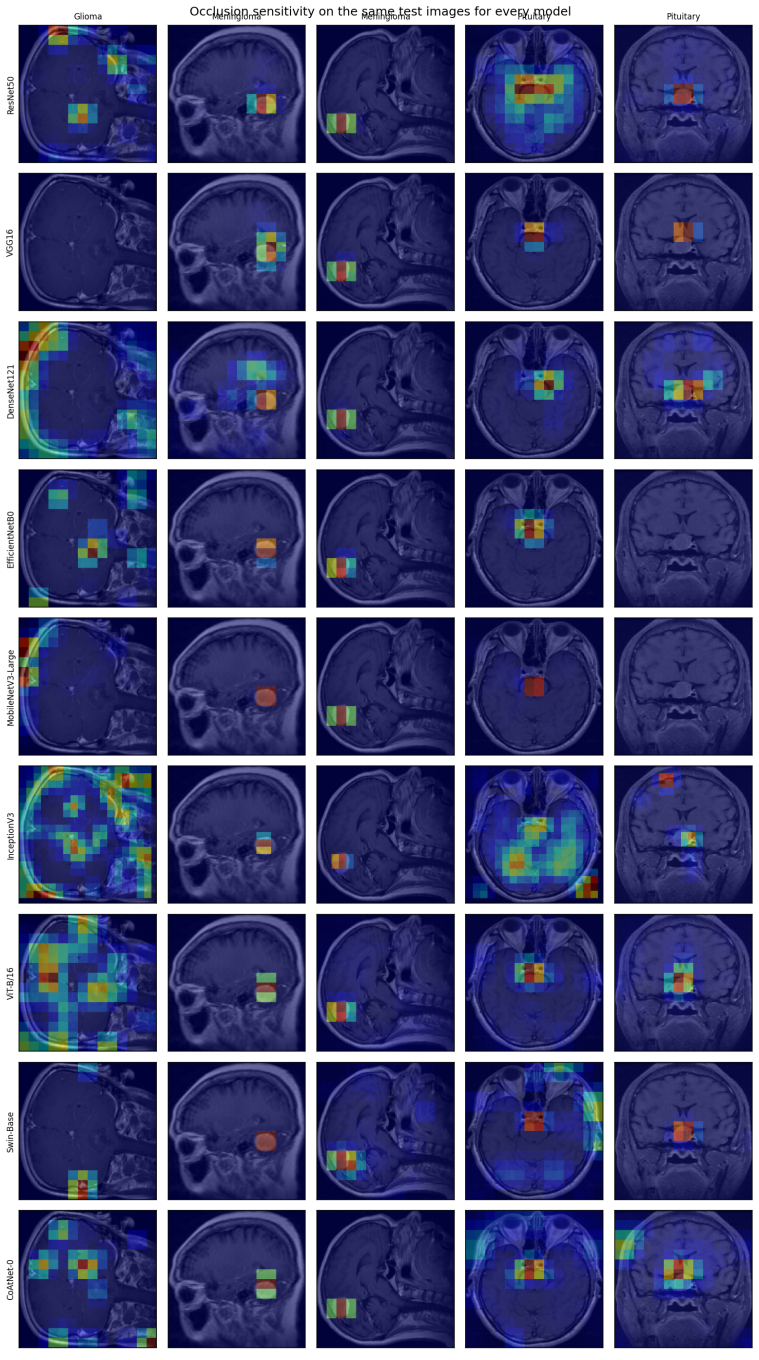



(b) Subtype classification.

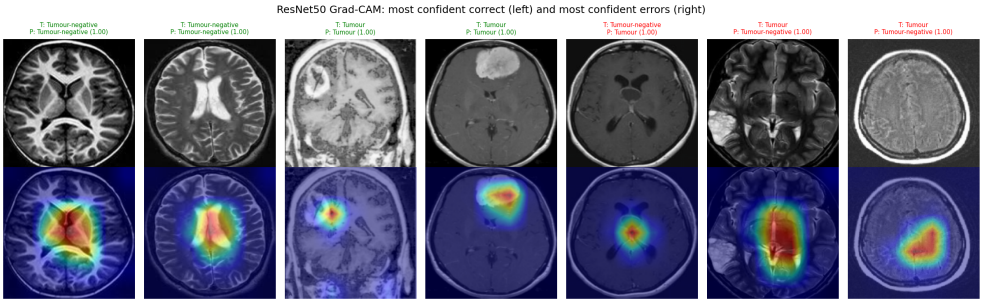
**Supplementary F14:** Share of Grad-CAM energy outside the head per model and class, the values of Supplementary T8. The black mark is the share of a uniform map.



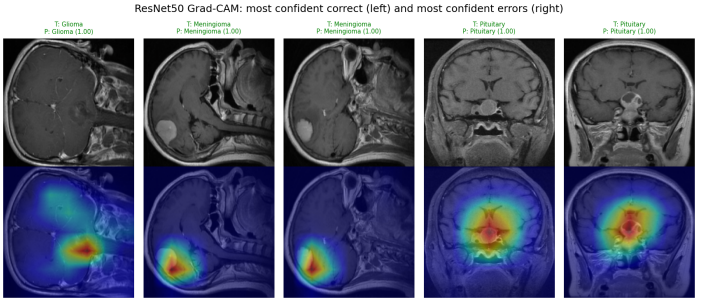

**Supplementary F15:** Binary detection: occlusion maps of every model on the same stratified test images.




**Supplementary F16:** Subtype classification: occlusion maps of every model on the same stratified test images.

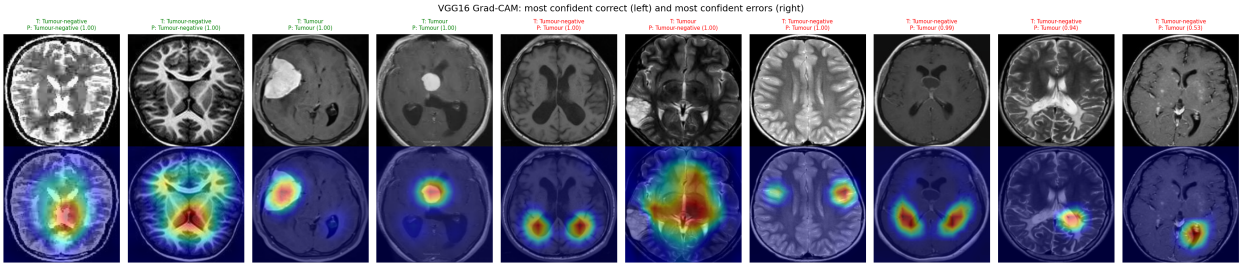



(a) Binary detection.

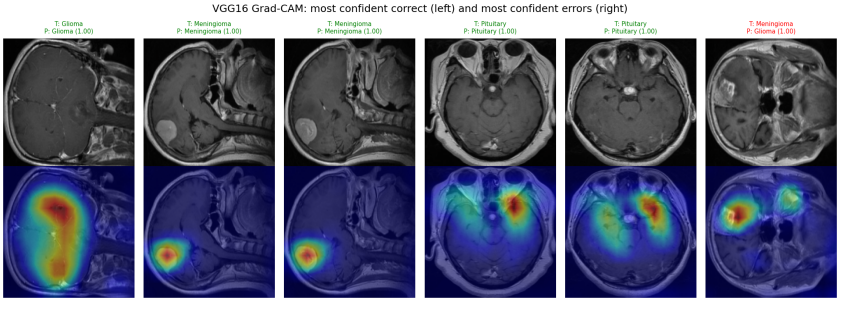



(b) Subtype classification.

**Supplementary F17:** Grad-CAM maps of ResNet50 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

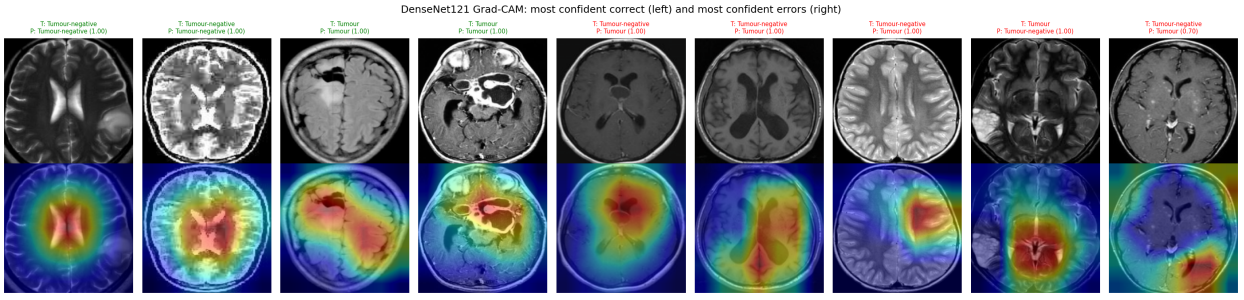



(a) Binary detection.

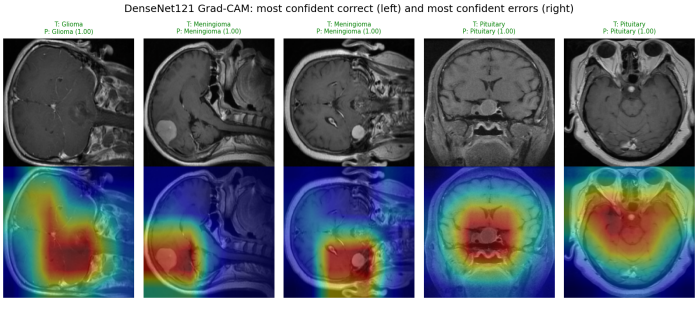



(b) Subtype classification.

**Supplementary F18:** Grad-CAM maps of VGG16 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

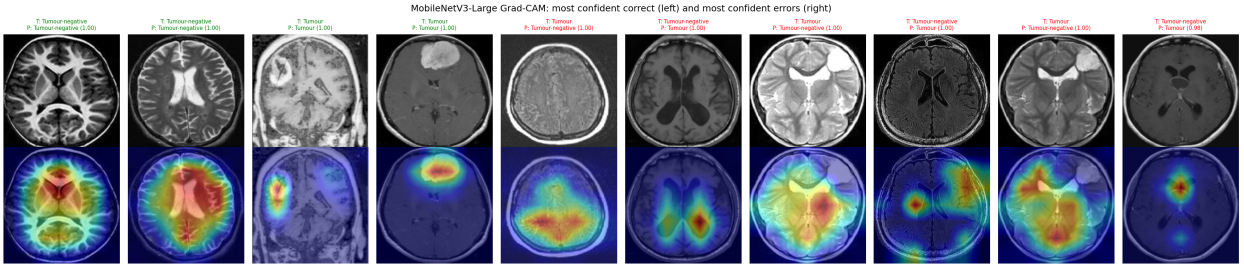



(a) Binary detection.

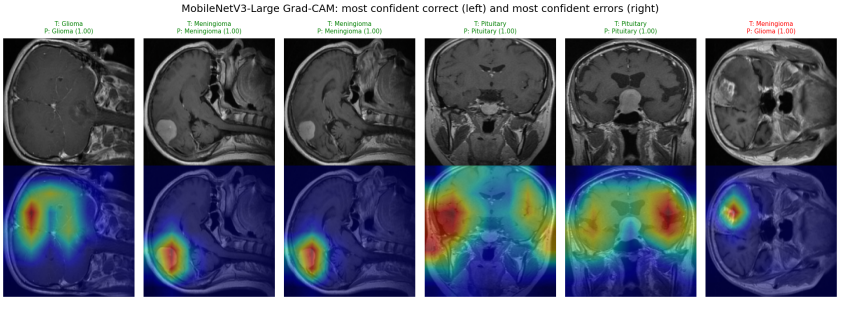



(b) Subtype classification.

**Supplementary F19:** Grad-CAM maps of DenseNet121 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

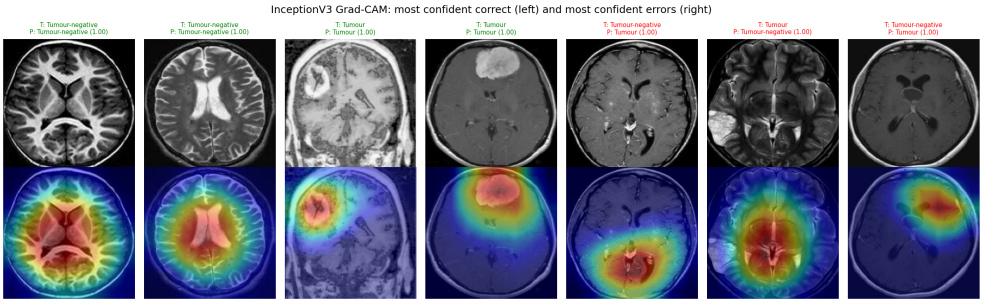



(a) Binary detection.

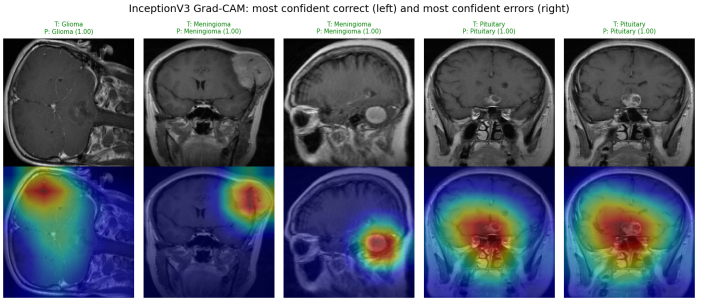



(b) Subtype classification.

**Supplementary F20:** Grad-CAM maps of MobileNetV3-Large on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

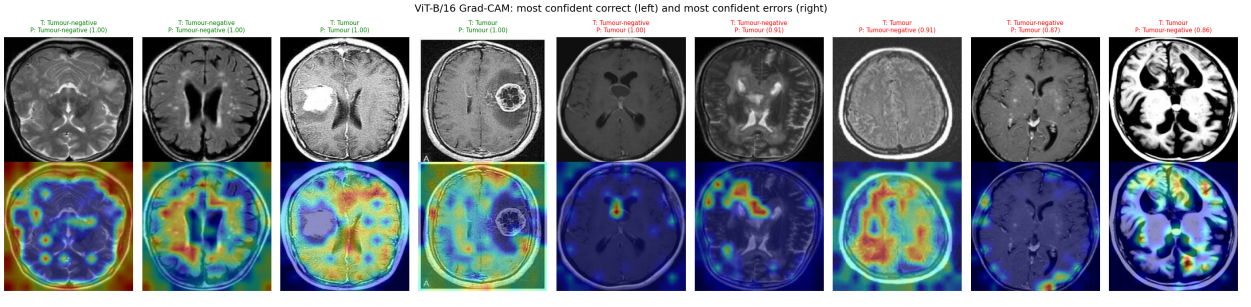



(a) Binary detection.

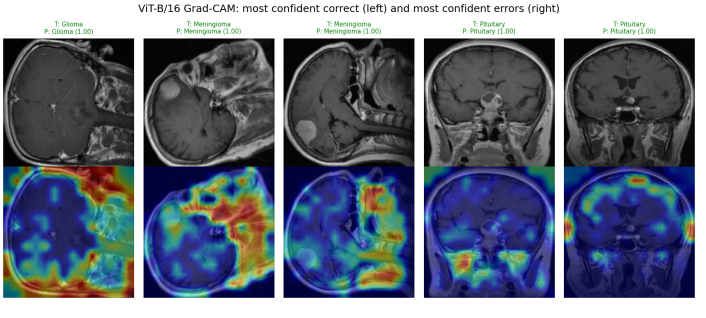



(b) Subtype classification.

**Supplementary F21:** Grad-CAM maps of InceptionV3 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

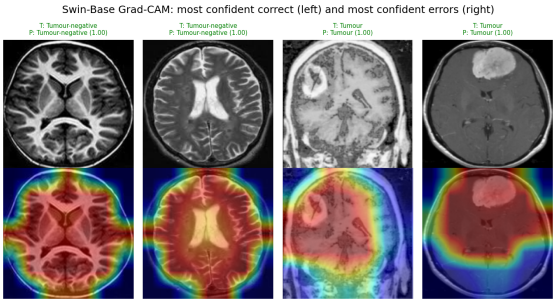



(a) Binary detection.

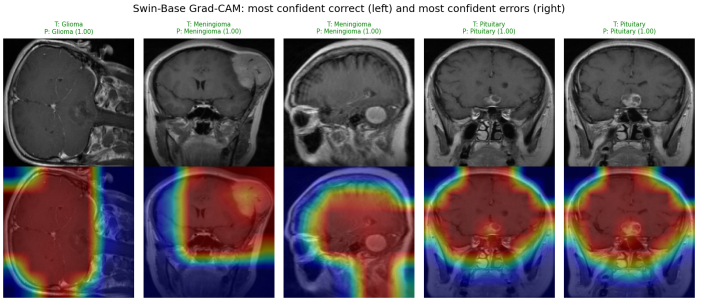



(b) Subtype classification.

**Supplementary F22:** Grad-CAM maps of ViT-B/16 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

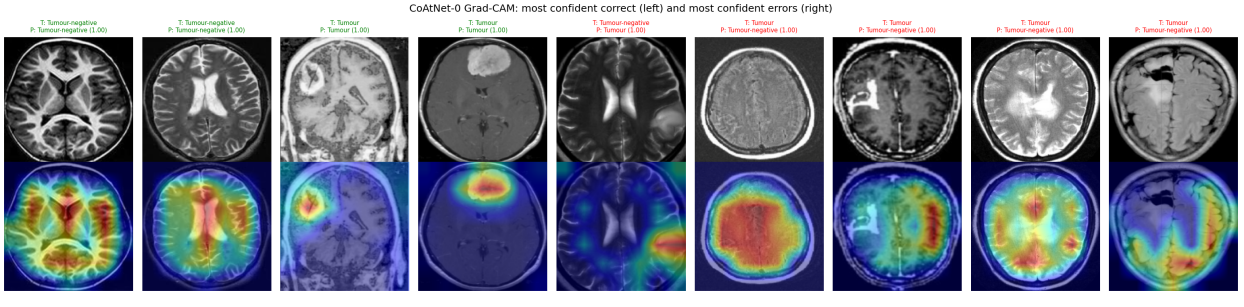



(a) Binary detection.

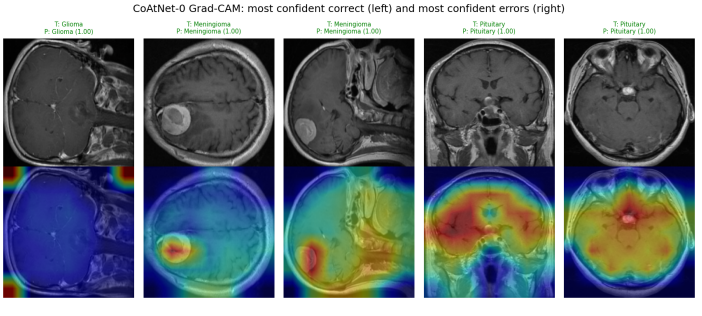



(b) Subtype classification.

**Supplementary F23:** Grad-CAM maps of Swin-Base on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

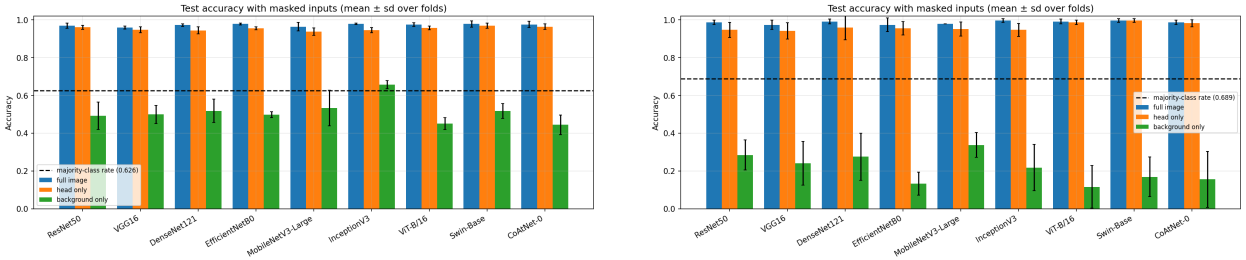



(a) Binary detection.




(b) Subtype classification.

**Supplementary F24:** Grad-CAM maps of CoAtNet-0 on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss, in the layout of Figure 3 of the paper. T: true class. P: predicted class and probability.

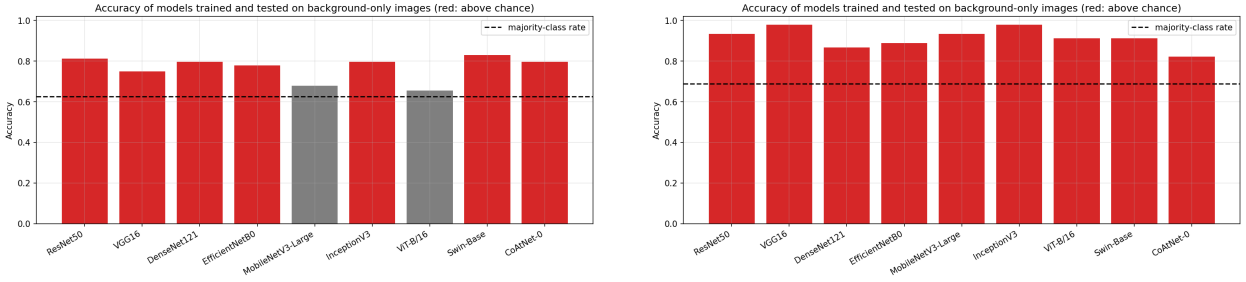



(a) Binary detection. (b) Subtype classification.

**Supplementary F25:** Masking test (inference only): test accuracy of each fold's checkpoint on full images, head-only images (background set to zero) and background-only images (head set to zero), mean ± standard deviation over folds.

**Supplementary T9:** Shortcut test: models trained (fold 0) and tested on background-only images. The p-value is a one-sided exact binomial test against the majority-class rate (0.6257 binary detection, 0.6889 subtype classification), Holm-corrected within each task.

| | Binary detection | | | Subtype classification | | |
|---|---|---|---|---|---|---|
| Model | Accuracy | AUC | Holm $p$ | Accuracy | AUC | Holm $p$ |
| ResNet50 | 0.8129 | 0.8657 | $< 10^{-4}$ | 0.9333 | 0.9968 | 0.0006 |
| VGG16 | 0.7485 | 0.8359 | 0.0013 | 0.9778 | 1.0000 | $< 10^{-4}$ |
| DenseNet121 | 0.7953 | 0.8578 | $< 10^{-4}$ | 0.8667 | 0.9904 | 0.0104 |
| EfficientNetB0 | 0.7778 | 0.8584 | 0.0001 | 0.8889 | 1.0000 | 0.0048 |
| MobileNetV3-Large | 0.6784 | 0.7821 | 0.1772 | 0.9333 | 1.0000 | 0.0006 |
| InceptionV3 | 0.7953 | 0.8606 | $< 10^{-4}$ | 0.9778 | 0.9984 | $< 10^{-4}$ |
| ViT-B/16 | 0.6550 | 0.7411 | 0.2396 | 0.9111 | 0.9717 | 0.0020 |
| Swin-Base | 0.8304 | 0.8893 | $< 10^{-4}$ | 0.9111 | 0.9976 | 0.0020 |
| CoAtNet-0 | 0.7953 | 0.8756 | $< 10^{-4}$ | 0.8222 | 0.9752 | 0.0337 |




(a) Binary detection. (b) Subtype classification.

**Supplementary F26:** Shortcut test: accuracy on background-only test images of models trained on background-only images (fold 0), the values of Supplementary T9. Red bars are significantly above the majority-class rate after Holm correction.